

ActInf Textbook Group ~ Cohort 5 ~ Session 19

(Chapter 8, Part 2) ~ 3/11/2024

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all right welcome cohort 5 it's March 11th and Andrew thanks for taking us on to chapter eight part two so go for it all right okay it's our second week on chapter eight um and just a brief refresher that this chapter is in the second part of the text book and so we're now focusing more upon building our own models and getting a sense of how to draw relationships between phenomena of interest and how to actually map them to variables in active inference models chapter 7 even we saw um discrete State space models and so we were looking at pdps and now chapter 8 now transitions to continuous State space models which is sort of the the foundation uh the starting point for uh historically for developing active inference models a lot of the earlier work of uh Friston and others was on continuous models first um so here we're looking at continuous State space models which are well suited for modeling let's say physical fluctuations uh impinging on sensory receptors and for the continuous motion of the effectors or muscles we use to change the world around us is a quote from the chapter um here we have model examples for dynamical systems involved in motor control as well as the concept of generalized synchrony with the bird song example and finally constructing Hybrid models uh near the end which would actually combine discrete and continuous variables um so it kind of almost add mixture of what we confronted in chapter 7 now with continuous variables being situated as well for example at a lower level of the model um since it's our second week uh from there I think we could just jump in does anyone have any thoughts uh on this chapter any questions anything they're they're still kind of uh crunching through uh in their mind about about how a lot of this works [Music] all I'll at one point thank you Andrew for the quick opening I think it's an interesting point that like you said it was initially uh with the earlier FR and Par models in continuous time and now the continuous time is being Revisited in the basic mechanic setting so it's kind of like a purely dynamical systems justification which is very similar to the SPM textbook so kind of classical equations of motion which kind of turns out to be like the um classical Newtonian basic mechanic so the kind of pre uh continuous time

which is where most of the this chapter's Focus is like including just the differential equations including the chaotic deterministic systems that's just the kind of pure dynamical systems approach again very similar to SPM and then this textbook doesn't really go into basing mechanics because this textbook was published March 2022 so probably was written between Frist's 2019 free energy principle for a particular physics and then obviously the book being published in so it's mentioned but many of the works that that made it clearer are only after the book just to kind of give a little more context on the continuous time yeah so we start with continuous we move to discrete and now we're back to continuous but it's kind of using what's been developed since then and also being able to yeah combine the two as well yeah in the the nested hybrid model at the end of chapter 8 is basically like picked up on by this paper which we discussed in live stream 46 and they basically pick up on that exact hybrid model I don't know if they use a figure let's see the hybrid 86 on page 166 oh sorry sorry yeah yeah you're looking at the paper yeah just that but I don't know they didn't include it there but uh here yeah figure 8.6 so here continuous time on the bottom because of the D derivatives and then discrete time on the top because of the transition Matrix and the explicit time and then in this paper they say okay with a belief desire intention like BDI model including discussion of like whether that is f psychology etc etc but within that commonly used like can you talk about beliefs desires and intentions and all these things they it it's not that it does better or any different even per se it just it's non-contradictory so that's kind of an interesting first way to put it hey oi is that particular paper does it discuss much like well I'm not sure how deeply you want to go into that or maybe another example but it it seems like a maybe a nice way of like finding um for week two here like um another application of what we find in chapter 8 like what's what's going on in that paper as far as like distinguishing discrete from continuous variables to have a recollection of like the model set up there let's look at another time but I think they all they mainly explored like differences in the belief desire intention map into the model and not sure how much they focused on that sensory motor part but yeah gotta sure yeah BR uh yes uh sorry uh could you please go to that uh sorry that desire part which they're mentioning uh in that paper the desire thing yeah can you please spend some time on this yeah I think in the you noticed it in the um the figure probably see yeah D AI they're using for discrete and that um M AI I think for motor AI so here they're looking at the um C preference distribution which is a distribution over observables and then um contrasting the observation of tasting ice cream and not tasting ice cream so these two um alternative ways you could parameterize that like in a program or an equation here's 7.3 like a

coin flip 7.3 70% of the time I expect to be tasting ice cream here it's 99% expecting to be tasting ice cream so then that c is stronger and so it entails a stronger motivational Force Through pragmatic value in E so that's one way to kind of look at it and then there's the other the way that the textbook kind of talked about it is with the c scaling but so depending on the program sometimes you're working in the scaled c um or unscaled scaled meaning to the statistical distribution or like here so that here Ice Cream versus no is being described summing to one like a coin flip like a real distribution so then you get or that could be represented as like 1 million to one so just depends on exact situation whether you're working in the kind of setting where you can learn by counting through experience or do other modifications and then when that's used as a distribution for pragmatic value it's transformed into something sus to one could you please send a link to this p paper uh in the chat box yeah it'll just be on the chapter 8 page yeah I'll put there too yeah so if there are no um anyone please feel free to bring up any other questions you might have along the way or you know toggle the reaction to raise your hand um but so in in this chapter along the way we're we're introduced to things like attractors um we're introduced as well to the concept of generalized synchrony comes up a bit later um and that's a particular notion that um you know as we know active inference agents are attempting to model the hidden States uh of the environment so they're attempting to model their environment what's going going on around them generaliz synchrony is this notion that um that the agent is basically their internal states are modeling or in certain ways beginning to synchronize with um the states around them we have no proof of course like the agent itself has no pure objective proof that it's somehow perfectly understanding the environment and all variables around it but nonetheless there's a kind of synchronizing going on and this can be applied to a kind of multi-agent setting and so here uh later on on in the chapter we're given this like bird song example which um you know the textbook is published in 2022 but the bird song example goes back to um at least 2015 potentially sooner um but the there are a couple 2015 papers by Friston and Parr where basically investigating using continuous variables um they simulate uh two song birds singing songs right and what they do is that um there's a we start to see this kind of emergent phenomena of course with with active inference models one of the big uh aims here of doing modeling is to like set up your variables set up your model run a simulation and then see what kind of emergent phenomena arise with the bird s song example they first um set up two birds that have identical models so not to not too uh surprising to find okay they have identical models they have the two song birds like quote unquote within earshot like they can the two

song birds have actions where they can sing and then they can take in sensory observations that being of continuous auditory signals and so it's like the birds are singing to each other and listening to each other the emergent phenomena that begins to happen is we start to see them taking turns it's as if they're singing from the same hymn sheet or the same script um and the generalized synchrony interestingly even occurs whenever they set up the model um the the simulation a second time around where the birds actually have different starting models so the birds start in different spots as far as their like State space like coordinates go and over time the bird's models actually begin through perceptual learning uh to to mirror one another so even though we start off with two different agents they start to synchronize um now they're singing the same hymn sheet again even though they're two you know very different birds as far as starting points go um the way that's modeled is through using attractors and a system and so uh in this case there's something that's called like a a synchronization of chaos which is rather interesting because you know with a chaotic system if you have two chaotic systems presumably they're you know even if you put them at the same starting point um their trajectories will very quickly diverge um you know with INF infin infinitesimally small chance that they will just magically line up but whenever you have two agents who are able to act um and listen uh to one another and they start to do this thing where they're following the same Hymn sheet suddenly you have a kind of synchronization of these two chaotic systems and we see those kinds of um Dynamics Illustrated in the in the chapter is kind of orbiting in figures 8.4 and 8.5 and um so yeah kind of a a lining up between different agents I think it's interesting for anyone who's curious about trying to set up systems or do multi-agent settings where they're trying to see you know how a joint action begins to play out between agents saying uh how agents can work together to um ultimately minimize their own distinct you know free energy um but by in order to do that they actually have to like yeah entrainment is a good word like one of them entrains the other um so that they can stay on the same page minimize prediction errors and it's fascinating because there are all kinds of uh Rich behaviors and dynamics that that occur um like the the the songs that are being sung presumably they like vocal Glides and brief pauses and so it's not just you know super simple Behavior that's that's being synchronized here but nonetheless it occurs see the bird song also I like how this chapter brings up oh uh Peres is that your hand up again uh yeah sorry I have a I may be wrong actually but just correct me um so for synchrony there needs to be some kind of a coupling uh I'm just wondering is it anywhere with the generative process and the generative models like the act like we have the like the Gen process is in a way uh

complementary to the generative model is it possible that that plays an impact sure so so in this example there's not too much going on in terms of U so just as a starting point so with modeling you're going to be defining a generative model and you're going to be defining a generative process in this example there's not too much distinct going on whenever a generative process is defined in fact here it's like we have two generative models so two agents and they're coming into contact with each other so it's like for bird number one its model is itself um its process that it's coming into contact with is just the other bird so you can think about the the observations that bird number one is taking in they're all being generated by by bird number two um as part of the the environment the process that's around it um does that make sense I think you you're right I think instead of thinking terms of a generative process Alto together like two different agents and they have like uh uh similar kind of internal model and uh and that is kind of causing kind of some kind of a coupling mechanism and so over a long period of time uh they may start to have behaviors which are similar to each other no like you mentioned that him sheet right so one person may say the starting lines and then then after sometime the second person will say the second thing then the third will say first will say the third and like that observations might happen yeah it's like the pendulum example which I think uh was mentioned last week like we have these two pendulums which are swinging and over a very very long period of time they'll they'll kind of um oscillate together no yeah yeah know that's exactly right and uh that's that's also referenced in the bird song paper so yeah just basically being able to start two different in that yeah in that case your pendulums are kind of like your agents um and yeah via you know via physics via the the the general logic that we're looking at here of like attractors yeah they begin to kind of move towards attracting points but they're acting as each other's attracting points so to speak um such that yeah their their behaviors start to to synchronize and so with the the birds the bird song is an interesting example versus the pendulums because the pendulums they just like continue moving for the birds um we have this turn-taking Dynamic that begins to occur because the the birds are trying to in terms of you know act act of inference and the free energy principle they're actively Trying to minimize their prediction error and the way they do that here is that they want to hear a song like they want to take in uh sensory observations that show that are realizing the preference of hearing a a song and the two ways they can do that that of course our perception changing their mind their own mind or uh action which is changing the world and so this turn taking Dynamic involves a combination of sensory attention and sensory attenuation uh is how it's phrased here and these are also um interesting

mechanics that we find in in just about any active inference model where action is involved um you have to so the birds take turns because like how do you keep a song going well one uh you need to hear that it's going on so they attenuate or or kind of like pause um their um their incoming sensory signals uh to uh commit actions that realize hearing the song it's like oh I don't hear a song I need to sing that's that's realizing my preference um okay I've sung um now I pause I hear the other bird singing I take that in as a sensory signal the other bird uh begins to sing maybe it sings notes that I expect great maybe it sings notes that I don't now we have prediction error starting to to jump up yeah and so whenever that happens of course the bird number one has to minimize its prediction error and so it starts to sing itself again and because these two birds are basically doing this back and forth there's a kind of synchrony where they're because they're both trying to minimize the prediction error the way that they sing the actions that they take the specific ways that they sing um like they start to synchronize they start to to match notes start matching expectations and that's the kind of thing that we see playing out in the the graphs um and the figures in the in the and the textbook um especially figure 8.5 on the right the authors here you know frisen and everyone they're trying to show uh like a before learning and after learning for these two birds on the left- hand side it's like okay these are the these are the Dynamics playing out here are the here are the dynamical equations um and then on the right you can see the upper graph before learning these are kind of like the state spaces of each Bird's expectations they're they top- down predictions um they're a little bit more all over the place whenever they start they begin to synchronize so that you know whenever whenever bird want and recall these are continuous um State spaces which is why we are able to do this as coordinates rather than having to look at like discreet um you know Point masses or something um so where bird number one's uh or excuse me yeah the first Bird's expectations in that lower right graph whenever they're around 10 um you know whenever we whenever we plot them um the second Bird's expectations are also around a value of 10 whenever bird number one's uh expectations or predictions are around 50 same case for the second bird so that's that's what they're trying to get at with these graphs is yeah they begin to the expectation even though they're dynamically shifting around we're never always at zero we could be at 50 this moment we could be at 5 this moment um the the important point is that the first Bird's predictions and the second Bird's predictions are um basically both lining up at just about any given moment one one interesting thing I hadn't seen or before was like here it's going between at least 60 but here it's only going to 40 yeah presumably I mean we have before learning and after learning so presumably they maybe just

before learning haven't explored quite as much of the this state space yet right like we have this much more EXP exploration after learning you've run more you know trials or at least extended the the time Horizon and then here they are like starting to syn up singing new notes that they'd never sung before or however you'd like to phrase that and yeah the the scale starts expanding because they're they're trying new things not sure nice nice interpretation yeah it's a it's it's a thought in any case and I also noticed it looks on the after learning graph it it looks slightly more chaotic or just messy not quite as synced up once you get past 50 on the after learning my guess is that like oh this is a new the new terrain that they're exploring here um yeah these little micro little micro densities yeah so presumably after more trials like we'll see an even finer kind of line or syncing up what happens between them we should find the at lab it might it may even be an SPM example there's also a kind of mini Standalone SPM like while it is a mat lab package there's a a kind of like compiled Standalone SPM oi or or others people like it it it's worth a a try yeah and the bird song I I I I could check or we could check together because it's kind of been around long enough to probably be included in the Dem demo yeah actually that's that's right um I was just having a look at the the paper um I think I dropped in the chat there but yeah just just about any of these uh friston co-authored papers prior at least prior to maybe the past couple years it's like going to a paper crlf SPM see if they you know have a script there um that they're you know they'll always mention in the paper of course there you go and um yeah it looks like it is in there um yeah for For Better or Worse of course matlab still requires um a license to use but but I myself have downloaded the the SPM Standalone before um you just you know unfortunately of course you cannot run M Matlab scripts if you don't have Matlab so spm's nice as far as like you get this um this like graphical system all of these different options of demos you can click you can look at the graphical outputs and yeah it looks like this one um is included in there it's like that um bird song uncore um oh no dmore demoore d.m so it can run M scripts in the background you just can't hop in and access them and modify them yourself uh to tweak the parameters or anything like that unfortunately but yeah it's nice it's there there like literally dozens of of preloaded scripts in there if you just download the SPM Standalone you just click one and uh you can you can watch what kind of graphical outputs it comes up with yeah and we can also explore bringing it into RX and F and everything yeah yeah definitely um and for anyone here who's not familiar um The Institute has an ongoing project Thursday mornings well sorry I'm in the US some ADV to that but um early on Thursdays uh there's an RX infer project for anyone who's interested in adapting the the logic of active inference that we're reading about here to direct

application in uh the Julia programming language and um right now we're going over different examples that are already provided by the developers byas Labs uh of this RX infer package um but um we'll be expanding from there uh different people you know are proposing their own Community projects uh you know diff different kinds of use cases are coming up with and that we could try uh emulating as a group uh and then another focus is that we would like to be able to reproduce some of these existing uh um projects or or topics for example here we could try and take the take the mat lab script for the bird song example and recreate it in Julia and uh finally just final note um is the um Julia programming language as well as the way that bias laabs is developing the RX infer package it's just it's nice that it's it's using this more um kind of sophisticated but also highly efficient method for the computation uh and message passing as we saw in the textbook back around chapter 5 um to to really just like being able to scale up these kinds of processes and and being able to just basically better use your computational resources save on some time um efficiency all that good stuff we in the programming World care about um yeah Francis you have your hand up uh yeah I just wanted if there's uh two questions actually firstly uh if there's likely to be any support for a pmdp group because I'm already having to Learn Python I I'm not I can't really commit to learning a second programming new programming language from scratch at the same time uh so I've been kind of slowly working through the um that nice pmdp tutorial um uh book Running it on the um is it collab uh and uh the other question is um is there anything to be learned or is it not a really helpful parallel uh between the idea of the bird song example as a kind of autocomplete and how uh does it does active inferences uh success here compare it uh in any way to um llms uh success at uh GPT for the world's smartest AO complete thanks just to quickly address it so right now like we do not have any active PDP specific group we could we could do a similar kind of thing right now the RX and fur is really it's it is kind of interesting it's it's um just from knowing PDP to medium degree it is different syntactically however like it is it's very different and it's a more General than pmdp in several ways not to say that that that one can't have other you know things the other doesn't have but just the that um RX and fur pulls way back like to the coin flip which was really helpful and then then builds up whereas PDP wraps it focuses on just kind of getting you to the discrete time active agent PDP as fast as possible because that's that's like not every use case but that's like many use cases and then it can kind of be modularized from there but RX and fur is it's a it's a very promising effort a lot broader so I I think it's a good way to go I'm kind of excited to learn it a lot ol is very good at it other than the second piece about the the um llms well one we we um

we haven't had a a streamer discussion on with these authors but this is one interesting paper recently um and then yeah like it's almost like a question of what what is perceived as a um Computing Benchmark especially if other considerations like interpretability or energy it's like is it a challenge is it a Computing advance to open a larger file with a computer with more RAM it's like yes but maybe it's not proportionally or differently a kind of benchmark so then what what are benchmarks for active models and like or or even kind of in another um setting like with the morph stream with Sarah Hamburg different guests it's been interesting to like hear how they talk about like the different characteristics of the biom mtic Computing system because they are not it's not simply just like the um CPU Ram descriptions thanks so much that's it looks like a really um interesting uh paper yeah and only six pages oh no comments of six pages yes yeah it's a sure it's definitely a starter so I and I think that they and many other people are interested and and a big um um like of course the models are different if you look at the Transformer architecture um or um but but there's some interesting with the state space models like the Mamba there's there's a few people like Matt Perkowski who have um talked a little bit about the relationship between the stat spaced models and active inference so that may be actually a closer relationship but um just in terms of how the Transformers are different than active inference models I mean in the textbook group chap these chapters are showing us what the ACM model is you can then just look at the Transformer architecture and just see how they are different architecturally but then um in in this paper they kind of highlight this kind of uh ongoing there there's various aspects to highlight there's the epistemic value there's the um reinforcement and but there's also a lot of overlaps as these are just large statistical models yeah oh thanks that's great oh and this is another good one where um they talk about how you in active inference you can construct the causal graph to be like that there is an uncertainty parameter or like a temperature parameter on some other parameter or there's like introspective access that using the um classic Lars live stream 28 figures with the meta awareness so if you construct the model to be this way then this parameter actually has that role in the system whereas if you ask a prompt no matter what you ask or how you ask it um to a a Transformer architecture and what is it like to be you and all these kinds of questions that people are very interested in it's only ever giving that semantic space fuzzy statistic of Words Answer so it could say I'm very sure that it's this way or very sure it's that way or I'm not sure or I feel this way when it's that way but those are all just words generated from texts um statistics of texts so they will never approach an actual introspection so here they raised the possibility of the actual introspection that's very interesting a totally different author list uh is

can we have that link on the um uh encoder as well thank you so much this is all interesting with the comparison to llms because I've been I'm no expert on that topic but I've been looking into it more myself and yeah there is this kind of um there's this kind of notion that okay llms are great as far as um you know kind of per performance goes with with accuracy and um and all of the computation there nonetheless yeah major the kind of criticisms and the sort of um debate or contrasting between llms and actm is just more generally like oh it takes a ton of computational resources to train in llm and then once you're kind of finished training it it's like okay well there it is it's not a it's not conceived as an agent who's um who's able to engage in action uh right so like try and it it's like to the idea would be you'd have to have an llm that had the capacity to actually go and explore information itself and then presumably it have you know from an active inference uh take it would have to have its own preferences of how to you know that it wants to realize so to speak or C Matrix in the or C vector and the PDP scheme or however you would like to think of it and um so it's it's the it boils down to the idea that an llm is passive um it can't go explore it can't update information and new you know expand its uh State space dynamically or or do things that maybe u a more active uh agent could um it's kind of like okay well once you've trained it like that's what you've got um nonetheless I'm very curious about the paper that yeah share because that looks like it it's it's actually saying no llms are not entirely passive but let's talk about it yeah that that that's all true and also what's really interesting though is the um I mean what what we might call ecosystems of shared intelligence or just the multi-agent prompting they like when you on perplexity AI or on um any internet enabled like cloud provider um or even if even in the uh local retrieval augmented generation where it's like you have a library of documents and then the it pre um assembles into the context window what's relevant like that's an information forging so this is kind of like um is it active inference like low road like has it been constructed according to chapter six using like forked code from an act in Freo or is it just being described as in in terms of having particular uh States like figure uh uh 9.1 with the kind of rat in the Maze like making a cognitive model of another language model that that doesn't mean it it will be useful to do that but the these models are prompting each other in ways that makes them have epistemic value like they do do web searches even if one language model locally doesn't have certain properties so there's a lot of um interactions which is kind of only to be expected and even these other these systems are are basically multi-agent models like like the large ones that are closed source but not in the kind of spatial Mobility way properly just like in the combining inputs gotcha I was actually personally not familiar with the fact that

like these are technically multi-agent setups um that's really interesting and also makes an immense amount of sense actually um yeah yeah and they're working off of one another another I'm not trying to over relate but just given this is the the textbook group of course it's like a you know we have the bird song example where the birds are trying to help each other um so to speak uh not we can't that's an interpretation to say that they're trying to help each other but the point being that they're both trying to minimize their prediction errors and so they're playing off of one another sharing information with one another or in this case a song guess with L llms in a certain sense of course them being constructed very differently we're talking about this right now but they're kind of working off of one another to yeah yeah I think another good uh point there in the context of building the models is like it's not necessarily about building like one Megatron agent that has every property it's about like what things are there and then starting to describe their interfaces which are usually less ambiguous and then describing the things and and their interfaces and interactions then maybe all you need is oscillator versus like a lot of times the kind of speculation about very very Advanced kinds of models but as we're on RX and fur like- which is awesome too the um the the the philosophy far outruns simple models so depending on what we're really trying to explore just you know describing and then having and then having the modularity of the model so we could start with a pendulum but then know that the interface would be described like in a stable way in RX and fur so that would have made large progress towards different kinds of phenomena without necessarily getting caught up in trying to locate it too early in a given architecture for a single agent can I do one more question does this idea of the uh synchronization um refer back to the phrase you used earlier about was it community of something a shared something yeah yeah I think here's here's a good citation okay good this one's it was on archive for some time then it was published here here's the ecosystems of intelligence or shared intelligence and this kind of lays out I guess other people can also give their summaries I see it it doesn't make any technical advances or descriptions there's only one figure um it's just laying out this idea of the interacting things through message passing this is the figure here's some manifold and then belief updating interesting reading format and then um then they just talk about how message passing amongst diverse intelligences in ecosystems of shared intelligence that's kind of the frame then they describe these um stages of development kind of an interesting read in terms of their timelines the scriptures Etc wow yeah thank you very much so it looks like in that paper they're describing uh like certain aspects of one agent's own model act as points of reference for other agents and vice versa so that you end up with this kind of like Ecology of

belief sharing a kind of collective intelligent intelligence resting on shared narratives and goals um again not to overrate everything here but with the bird song example I had actually like I this past week took some time to like actually read that paper itself the bird song One not just its briefer description uh in the textbook and there's a lot of there's a lot of interesting things actually going on in that 2015 bird song paper I in some ways this might be just designing ecosystems of intelligence from first principles more than likely has some relationship there because the bird song example it's not just of course it's not just about trying to talk about distinctly birds and and song birds and like you know uh as as interesting as that could be from a biological or Zoological or otherwise standpoint it's really trying to get at something like how do agents share information with one another and kind of in Trin one another how do we have how and why do we see things like convergence of beliefs um as far as biological organisms go we can see a synchronization of things beyond the words um we can see like whenever two people are talking to each other their the rate of their speech tends to begin to converge over time or or exhibit you know near identical characteristics um the kind of vocabulary that people use tends to kind of sync over time as well um and then yeah there's so there's this kind of shared narrative a kind of um you know from a seal agent perspective uh the single agent is starting to model um the behavior of the world around it the process as well as its own um two agents might start to model each other and in a sense uh there uh the bird song paper claims it's not just that once you get to this multi-agent setting where where the agents are entraining one another and starting to show identical behaviors and internal model mappings and U modeling of the world of each other uh it's not just that they're modeling the other Bird's Behavior or their own behavior they phrase it as they're beginning to like our Behavior like there's a kind of collective behavior on the on the horizon here that they're starting to model um yeah I I just think that's a very very interesting and potentially one of the core aspects of trying to to get into this kind of collective intelligence synchronization kind of logic here yeah yeah great point I mean when it's on a onedimensional manifold like in the extreme case or even just as a heris like if 80% of the variability is captured by one number like you're on a a one-dimensional manifold on the two-dimensional space that's the that's what a manifold is and then yeah you're totally right this is not really a uh it's not an Ornithology paper in the traditional sense because it is about hermeneutics and it is with this other joint paper that they reference that's like a a twin paper and so it it's it's totally interesting that it has a lot to do like you said with narrative and with uh interpretation here they um they take like a dynamical systems in approach to interpretation later that is also like um expanded

and complemented by the quantum reference frame Chris Field's syntax and semantics work but this is like a perception action grounded dynamical hermeneutics so not addressing all semantic or Semitic questions but that but like how dynamical systems can have General synchrony which doesn't mean doing the same thing it can mean it can mean Mutual information increases through differences like or in um in the Chris Fields setting like the the principle of unitarity and like that systems um come into entanglement with each other and then that was discussed like does that mean that they go to the same States like no not necessarily and I say as a counselor who goes into conversations with people where they know their world but maybe they're stuck and I don't know their world but I have models of how to get unstuck this is uh wow um very very very interesting thank you yeah no super super interesting I mean I think I I think there's enough to go off of here at least theoretically that potentially one of the key um aspects of Counseling and as well as other things clinical therapy and so on is that uh just basically a so-called talking cure or some s some such thing is for someone who's having uh kind of repetitive uh you know behaviors or thoughts or you know whether you want to view them from a symptomatic perspective in the cases of like I don't know depressive rumination or a variety of other kind of things that come up like in one sense from an active inference perspective it might be that someone is having something like difficulty with belief updating and that by talking like actually just talking with another human being um you might see the person who's who's you know the client you might start to see them like you know not not to be too theoretical or kind of philosophical here but there's almost a kind of potential s thinking to some degree of what we might call their internal States or their behaviors or otherwise with that of the counselor or the the therapist you know um whole different set of questions and and research to be done or research that's already been done just try and claim like that that can have a a long-term impact of course like counseling or therapy like there are regular sessions so you're getting like repeated instances of these these temporary sinking EP uh sinking Up episodes going on and then another aspect of course is that the counselor and or therapist they don't want to actually themselves imitate um the behaviors being seen by the client they want to help not match um but yeah just you know just trying to extrapolate a little bit here yeah that's why we have a a fairly precise definition of empathy as if we were feeling the same as our client as if we were in their world but not actually being in their world I have to leave thank you so much all right sure next time fors chapter nine thank you Andrew so see you all next time bye take care